

Samson Swaroop Paturi

Lawrenceville, GA 30044 | 6468815553 | swaroop.samson@gmail.com

Summary

AI, Data Product, Project & Platform Leader with 10+ years of experience building and scaling infrastructure-grade AI systems and autonomous platforms. Bring rich experience building and scaling mission-critical AI/ML platforms. Expert in **Agentic AI**, **MLOps**, and **Developer Tools**, with a track record of moving products from "Experimental Demos" to "Industrial-Grade Infrastructure." Specialized in hardening **RAG architectures** and **Reasoning Loops** to ensure 99.9% reliability in high-stakes environments. Proven track record of owning end-to-end software roadmaps for Agentic AI workflows, having successfully implemented automated pipelines for 100,000+ distributed endpoints to drive a 40% reduction in manual triage effort. Expert in leveraging LLMs, LangChain, and RAG services to transform complex, unstructured telemetry into actionable diagnostic insights and executive-level summaries. A strategic leader who has delivered ~\$1.5M in annual platform savings by centralizing analytics capabilities and eliminating redundant toolsets across global business units. Passionate about bridging the gap between Data Science, Engineering, and GTM teams to turn early-stage AI pilots into hardened, AI-native platforms that improve uptime and deliver gigawatt-scale operational impact in highly regulated environments.

Skills

- **AI-Native Product Ownership:** Lead the end-to-end lifecycle for AI and Agentic AI products, focusing on autonomous workflows and human-in-the-loop systems to drive R&D efficiency.
- **Project/Program Management:** Apply OKRs to align project and program roadmaps with business strategy, ensuring AI deployment directly influences churn reduction and expansion revenue.
- **Executive Stakeholder Engagement:** Act as a trusted advisor to leadership, aligning AI initiatives with business strategy and leading vendor evaluations for optimal cloud services and AI tools.
- **Cross-Functional Orchestration:** Facilitate seamless collaboration between data science, engineering, and GTM teams to translate business friction into high-value AI products.
- **Data-Driven Insights:** Analyze structured and unstructured customer data using Python, R, and statistical modeling to identify trends, root causes, and actionable insights.
- **Operational Excellence:** Partner with operations to optimize AI/ML pipelines, focusing on performance monitoring, proactive issue detection, and automated alerting.
- **Agile Visualization & Governance:** Utilize Miro, Jira, and Confluence to drive architectural clarity, document reusable assets, and maintain governance across the AI lifecycle.
- **Technical Depth:** Leverage expertise in AWS (SageMaker, S3, Redshift), KubeFlow, and MLflow to ensure platform reliability, scalability, and technical feasibility.

Highlights – AI Product Platform Technology Leadership

- **Global Team Leadership (Engineering, Analytics, Governance, DS):** Led high-performing global teams across multiple time zones, fostering a culture of ownership, innovation, and disciplined execution. Scaled capabilities by developing future-ready talent, establishing cross-team collaboration frameworks, and improving delivery velocity across distributed engineering and analytics groups.
- **Shared Analytics and Cost Efficiency:** Centralized insight-generation capabilities used across multiple business units, eliminating redundant tools and achieving ~\$1.5M in avoided annual platform spend.

- **AI, GenAI & Agentic AI Enablement:** Championed enterprise AI strategy by integrating GenAI and agentic workflows into data platforms, accelerating automation, insight generation, and operational decision-making. Defined governance, ethical use standards, and scalable enablement patterns to responsibly adopt AI across business units.
- **Data Governance, Metadata, MDM, Quality:** Established enterprise governance frameworks that strengthened trust, compliance, and operational consistency across regulated markets. Standardized metadata, lineage, and master data practices to enable global interoperability and ensure high-quality, decision-ready data.
- **Enterprise Modernization & Automation:** Led transformation programs that simplified fragmented environments, introduced automation across ingestion and quality processes, and unlocked significant efficiency gains. Built modernization roadmaps that balanced innovation with operational stability, resulting in faster delivery cycles and reduced technical debt.
- **Executive Communication & C-suite Stakeholder Management:** Advised senior executives on data strategy, AI readiness, and transformational investment priorities. Distilled complex technical topics into clear, strategic narratives that influenced decision-making and secured enterprise sponsorship for multi-year data initiatives.

Experience

SR PROJECT AND PRODUCT MANAGER – DATA STRATEGY | 02/2023 – 01/2026

Capgemini (National Grid)

- **Product Vision:** Defined the roadmap for an **Agentic AI platform** that disrupted traditional "Detect-and-Respond" workflows, moving the system toward **Autonomous Investigation** and reducing human cognitive load by **40%**.
- **ML & LLM Hardening:** Spearheaded the integration of **RAG (Retrieval-Augmented Generation)** and LLM reasoning loops; personally audited **5,000+ reasoning traces** to implement **"Chain of Thought" transparency**, successfully eliminating hallucinations in critical outputs.
- **Engineering Velocity:** Built internal-facing data platforms and **MLOps pipelines** for global engineering teams, enabling rapid experimentation and deployment of predictive models across **100,000+ edge nodes**.
- **Operational Efficiency:** Acted as a "momentum catalyst" during a 6-week "Zero-to-One" sprint, manually re-structuring 400-page legacy technical manuals into **AI-optimized markdown** to ensure high-fidelity retrieval for autonomous agents.
- **Facilitated executive steering reviews**, communicating platform adoption metrics, model performance, deployment readiness, and ROI from AI-driven automation initiatives.
- **Commercially Focused AI:** Integrated LangChain and Azure OpenAI RAG services to support executive summaries and diagnostic insights, consumable across diverse business domains.
- **Governance & ROI:** Established platform-wide standards for GenAI-driven analytics and model usage guidelines, ensuring measurable ROI and consistent adoption across the enterprise.
- **Agile Delivery:** Managed a shared backlog for ETL and ML services, delivering five major releases on time with a 25% improvement in team velocity
- **Implemented agentic AI pipelines** within the platform to automate ETL metadata generation, validation, anomaly scoring, and quality monitoring across 100,000+ smart meters, improving reliability, standardization, and repeatability for multiple consuming teams.
- **Built cross-suite reusable components** by collaborating with Product, Engineering, and Data Science to centralize natural-language data exploration, enabling multiple business units to use GenAI-enabled assistants for AMI insights, without building parallel tools.
- **Integrated LangChain and Azure OpenAI RAG services** into the core platform to support shared use cases, such as automated reporting, executive summaries, diagnostic explanations, and stakeholder insights, consumable across business domains.
- **Established platform-wide standards and governance frameworks** for GenAI-driven analytics, metadata quality, KPI definitions, and model usage guidelines, ensuring consistent adoption, and measurable ROI across internal teams.

IT WF INC (DATA SCIENCE PROJECT MANAGER) | 11/2021 - 01/2023

Duke Energy

- Led end-to-end development of the **Customer Mapping Engine (CME)**, an ML-driven platform designed to correct inaccurate grid connectivity mappings by predicting **meter-to-transformer** and **meter-to-phase** relationships, improving enterprise data accuracy from **~60% to 92%** and strengthening downstream grid reliability analytics.
- Owned the product and technical roadmap for CME, translating distribution operations pain points into a scalable AI/ML platform integrating AMI telemetry, GIS connectivity, and asset datasets.

- Designed and operationalized machine learning pipelines leveraging **K-Means clustering**, **KNN similarity modeling**, and **XGBoost supervised classification** to infer transformer associations and phase connectivity with confidence scoring.
- Led delivery planning and execution for the Customer Mapping Engine platform, managing scope, sprint milestones, and cross-team dependencies across data science, GIS, and grid operations.
- Owned backlog prioritization and roadmap execution for connectivity intelligence capabilities, aligning mapping corrections with enterprise grid modernization initiatives.
- Coordinated data acquisition and validation efforts across AMI, GIS, and asset systems, resolving integration blockers and data-quality dependencies.
- Led feature engineering initiatives across AMI interval data, deriving load-shape signatures, voltage behavior patterns, transformer neighborhood aggregates, and correlation metrics to strengthen model signal quality.
- Established standardized training datasets by integrating SCADA/AMI data with GIS records and maintenance systems, implementing label validation, taxonomy alignment, and feeder-level segmentation for robust model training.
- Built production MLOps pipelines using cloud infrastructure to automate data ingestion, feature extraction, model training, evaluation, and batch inference publishing to downstream GIS systems.
- Implemented model lifecycle governance including experiment tracking, version control, performance monitoring, and drift detection to ensure sustained mapping accuracy and operational trust.
- Directed backlog execution and cross-functional delivery across data science, engineering, and grid operations teams, aligning model outputs with business workflows and reliability planning systems.
- Coordinated stakeholder engagement across operations, asset management, and IT teams to drive adoption of ML-generated connectivity corrections within enterprise grid platforms.
- Enabled automated confidence-based publishing workflows, prioritizing high-certainty corrections for direct system updates while routing lower-confidence predictions for SME review.

Capital One Data Project Manager (Short-term Engagement | 3 Months)

- Led rapid discovery and MVP definition for a cloud-based data product supporting credit risk and customer analytics; established product vision, success metrics, and prioritized backlog aligned to underwriting and marketing use cases.
- Partnered with Risk, Compliance, Data Science, and Engineering teams to translate regulatory and analytical requirements into governed data features, canonical models, and secure access patterns (PII masking, RBAC).
- Collaborated with data platform teams (AWS/Snowflake) to streamline ingestion, data quality, and observability frameworks, improving trust and usability of downstream analytics datasets.
- Delivered MVP within a compressed delivery window using Agile/Scrum ceremonies, enabling faster analyst access to curated data and reducing manual data preparation effort.

DATA SCIENCE PRODUCT OWNER | 09/2019 - 11/2020

Florida Power and Light

- Managed predictive maintenance program execution for wind turbine component failure detection, aligning reliability engineering, field operations, and data science delivery workstreams.
- Led requirements gathering workshops with turbine technicians and generation teams to define model objectives, failure taxonomies, and operational alert thresholds.
- Coordinated cross-site pilot deployments across remote wind farms, managing alert integration, maintenance scheduling alignment, and adoption readiness.
- Led the end-to-end product strategy for "WindSentinel," a prescriptive O&M platform managing North America's largest wind fleet, transitioning operations from reactive "run-to-failure" to data-driven Condition-Based Maintenance.
- Engineered a billion-dollar risk-mitigation roadmap by narrowing product scope to the "Catastrophic Three" assets (Gearboxes, Generators, Main Bearings), accounting for 70% of unplanned CAPEX.
- Orchestrated the development of an MLOps pipeline capable of ingesting and harmonizing 1TB+ daily of heterogeneous data, including high-frequency SCADA telemetry, controller event logs, and unstructured technician maintenance records.
- Improved model precision by 30% through a Human-in-the-Loop initiative, digitizing "tribal knowledge" from field technicians to create high-integrity training labels for vibrational and thermal anomaly detection.
- Delivered \$2.4M in avoided CAPEX within the first six months by predicting gearbox failures weeks in advance, allowing for "up-tower" repairs and eliminating emergency heavy-lift crane mobilization fees (\$150k+ per event).

- Reduced fleet-wide unplanned outage time by 18% by implementing a "Health Index" scoring system, allowing reliability teams to optimize logistics and spare parts inventory months ahead of predicted failures.
- Bridged the gap between Data Science and Grid Operations by implementing "Reasoning Transparency" in the product UI, translating complex model outputs into actionable "Prescriptive Repair Orders" for non-technical field crews.

SENIOR DATA SCIENTIST | 08/2017 - 09/2019

Comtec Info (NextEra Energy Resources)

- Managing vegetation and tree growth around power lines is an important responsibility of utilities.
- Historically, utilities have approached veg mgmt using a cyclical, time-based approach—inspecting and trimming the areas around their lines periodically.
- This project aims to move away from a schedule-based maintenance routine to a more predictive/prescriptive solution.

PROJECT MANAGER | 06/2006 - 08/2017

IBM

Contact Center Operational Excellence (IBM/AT&T)

- Omni-Channel Strategy: Orchestrated global support workflows across multiple digital channels, leveraging predictive staffing models and real-time traffic analysis to maintain consistent 80/20 service levels.
- Operational Optimization: Scaled asynchronous support models and engineered tiered queuing systems that reduced First Response Time (FRT) by over 80% and Average Handle Time (AHT) by 15%.
- Data-Driven Process Improvement: Utilized rule-based sentiment analysis and keyword-mining to identify systemic technical friction, driving a 15% increase in First Contact Resolution (FCR).

GRADUATE ENGINEER | 05/2005 - 06/2006

Dell Technologies

Education

Indian School of Business - Certificate Program

Advanced Management Program in Business Analytics, 01/2015

JNTU - Bachelor's Degree

Electrical, Electronics and Communications Engineering, 01/2005

Websites, Portfolios, Profiles

- [linkedin.com/in/samson-swaroop-paturi-49475625](https://www.linkedin.com/in/samson-swaroop-paturi-49475625)

Licenses & Certifications

- Six Sigma Green Belt - RABQSA
- Advisory Product Services Professional - IBM
- ITIL Foundation Level - AXELOS Global Best Practice
- Certified Scrum Product Owner (CSPO) - Scrum Alliance
- SAFe Product Owner/Product Manager – Scaled Agile Framework Inc
- Certified Scrum Master - Scrum Alliance